

Statistical Inference for Agent-Based Epidemiological Models: Enabling Real-Time Disease Forecasting

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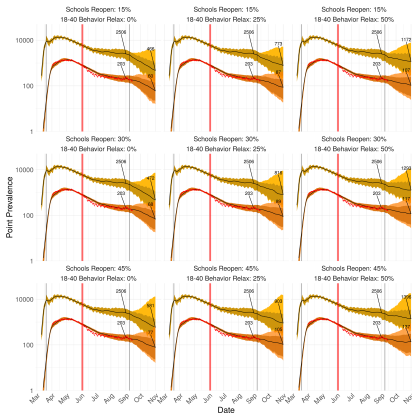


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Motivation

School reopening scenario in Chicago



- Prediction and forecasting of disease spread
- Support timely public health decisions
- Improve understanding of complex interventions

Agenda

- **Epidemic models**

- Brief overview of Agent-Based Models (ABMs)
- Strengths and challenges

- **Statistical Inference and Calibration**

- Need for efficient parameter estimation
- Bayesian optimization for ABM calibration
- Computational and inferential challenges



What is an Agent-Based Model?

- Individuals explicitly modeled as agents
- Agents interact based on simple rules
- Emergent population-level dynamics
- Effective for modeling interventions



Source: <https://www.bankofengland.co.uk/quarterly-bulletin/2016/q4/agent-based-models-understanding-the-economy-from-the-bottom-up>.



Building an ABM for Infectious Disease

Core components

- **Population Structure**
 - Demographics, contact networks
 - Mobility and spatial interactions
- **Disease Transmission Model**
 - Infection rules (e.g., susceptible-exposed-infectious-recovered)
 - Transmission probabilities and contact patterns
- **Behavioral Rules**
 - Agent movement and interactions
 - Response to interventions (vaccination, isolation, social distancing)



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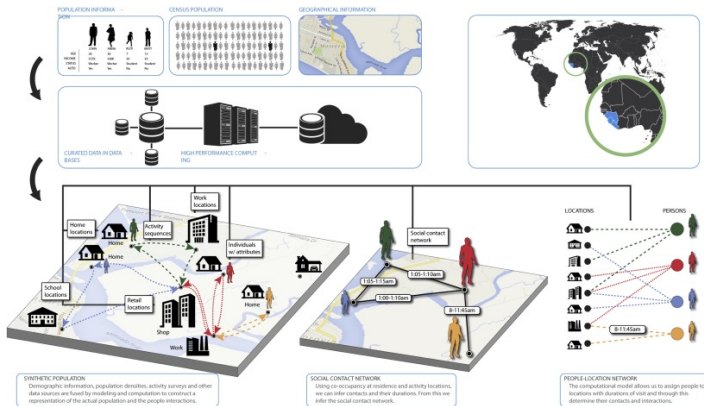


Image courtesy: Henning Mortveit



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Example

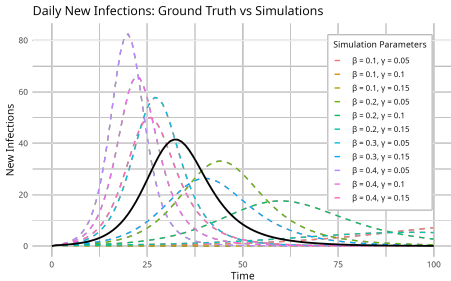


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Fitting the model

- This is a simple model with two key parameters:
 - β : the transmission rate
 - γ : the recovery rate
- To use this model for prediction or forecasting, we need to estimate (β, γ) from data.



Fitting the model

Estimating (β, γ)

- **Random Search:**

- Try many combinations (β, γ)
- Simple, embarrassingly parallel

- **But...**

- ABMs are expensive to run
- High-dimensional parameter space
- Limited computational budget

- **We need to be smart:** How can we explore the parameter space *efficiently*?



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Surrogate models for fast inference

- **Challenge:** Agent-based models (ABMs) are expensive to simulate – can't afford to evaluate them thousands of times during inference.
- **Solution:** Build a **surrogate model** – a statistical or machine learning approximation of the ABM output.
- Once trained, the surrogate is:
 - Fast to evaluate (milliseconds)
 - Differentiable (if needed)
 - Uncertainty-aware (e.g., Gaussian Processes)
- We can then use this surrogate within a Bayesian inference framework to estimate ABM parameters efficiently.



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Gaussian processes for surrogate

- Many surrogate modeling choices exist: Neural networks, random forests, gradient boosting, etc.
- **Our preferred surrogate is Gaussian Processes (GPs)**
 - Non-parametric flexibility
 - Quantified prediction uncertainty
 - Closed-form posterior inference
- A Gaussian Process is a distribution over functions, such that

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$$

where:

- $m(\mathbf{x})$ is the mean function
- $k(\mathbf{x}, \mathbf{x}')$ is the covariance (kernel) function



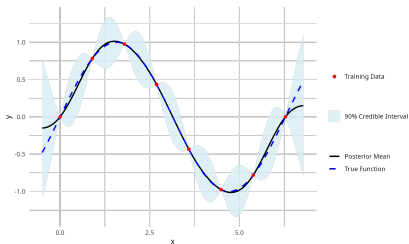
Gaussian processes for surrogate

- For a finite set of inputs $\{x_1, \dots, x_n\}$, the outputs $f(x_i)$ are jointly multivariate normal:

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- For a finite set of inputs $\{x_1, \dots, x_n\}$, the outputs $f(x_i)$ are jointly multivariate normal.
- The GP posterior at new points x^* is:

$$f(x^*) \mid \mathbf{y} \sim \mathcal{N}(\mu(x^*), \Sigma(x^*))$$



GP fit to observations of a sine function. The shaded region shows the 90% credible band.

Bayesian Optimization

Bayesian Optimization (BO) is a global optimization method for expensive black-box functions. It uses a surrogate model to guide efficient exploration of the parameter space.

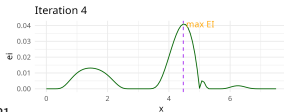
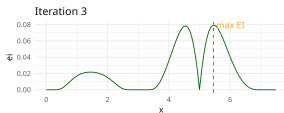
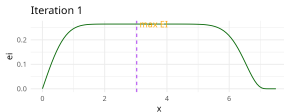
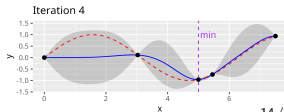
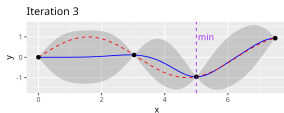
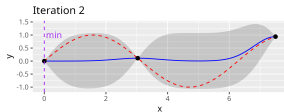
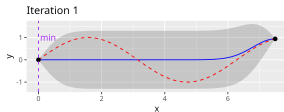
Pseudocode:

- 1: Initialize with a small set of input-output pairs $\{(x_i, y_i)\}_{i=1}^n$
- 2: **while** evaluation budget not exhausted **do**
- 3: Fit surrogate model $\hat{f}(x)$ to current data
- 4: Maximize **acquisition function** $a(x; \hat{f})$
- 5: Evaluate simulator at new point $x^* = \arg \max a(x)$
- 6: Augment data: $\{(x_i, y_i)\} \leftarrow \{(x_i, y_i)\} \cup (x^*, y^*)$
- 7: **end while**
- 8: Return best x observed



Bayesian Optimization

A working example



Putting it all together

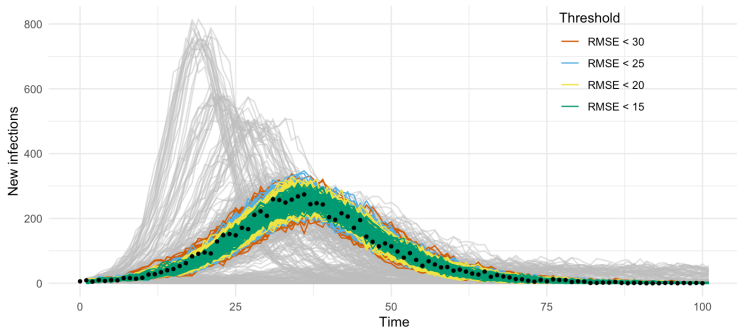
What we have so far

- **Agent-Based Models (ABMs)** are powerful tools for modeling complex infectious disease dynamics at the population level. But they need calibration.
- **Gaussian Process (GP) Surrogates** provide a principled way to approximate the expensive ABM outputs, with uncertainty quantification.
- **Bayesian Optimization (BO)** leverages GP surrogates to efficiently explore the parameter space by balancing exploration and exploitation.
- **Next:** We apply these tools to calibrate ABMs for real infectious disease data (e.g., COVID-19, flu, Ebola).



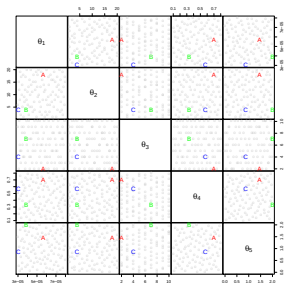
Application

Toy example

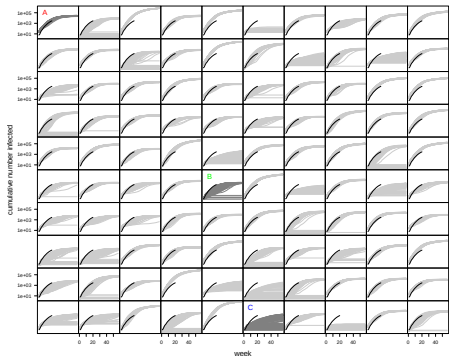


Application

Ebola in West Africa - EPIFAST model



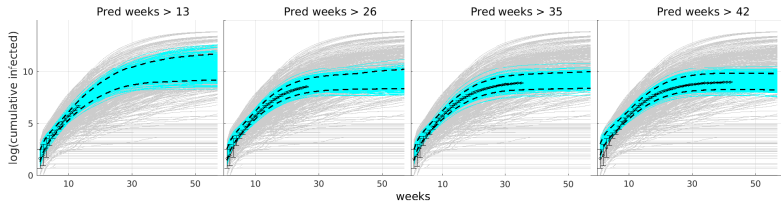
A design of size 100



Simulation output at 100 design points and
observed data

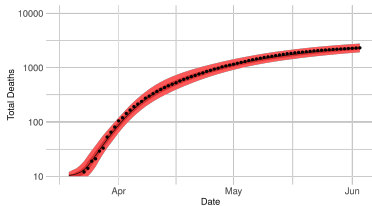
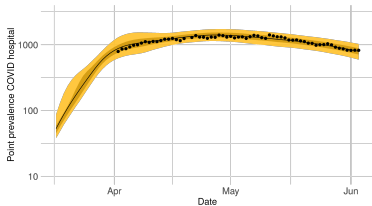
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Application

Covid19 in Chicago - CityCOVID model



What We Didn't Cover (But Should)

- **Stochasticity and Uncertainty:**
 - Handling stochastic ABM outputs in inference
 - Incorporating uncertainty into parameter estimates and forecasts
- **Scalability and Complexity:**
 - Multi-output calibration for multiple surveillance targets (cases, deaths, hospitalizations)
 - High-dimensional parameter inference and surrogate model scalability
- **Workflow and Real-Time Adaptation:**
 - Design of experiments to select informative simulations
 - Adaptive inference with streaming real-world data



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References

- Fadikar, A., Higdon, D., Chen, J., Lewis, B., Venkatramanan, S., & Marathe, M. (2018). Calibrating a stochastic, agent-based model using quantile-based emulation. *SIAM/ASA Journal on Uncertainty Quantification*, 6(4), 1685-1706.
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